

Revised Generation Interconnection System Impact Study Report

For Queue Project Q12-0126

East Bismarck 100 kV
January 27, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0126. The Interconnection Customer proposes a 80.0 MW wind facility interconnecting at the East Bismarck 100 kV substation in the Illinois area. The request was received on April 21, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 89 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,403,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Terminus / Mtn Island 100 KV
Contingency: Texon / E Bismarck 230 KV L
Loading: 144.5% (Rating: 98 MVA)
DFAX: 37.8%
Pre-project loading: 113.4%
2. Facility: Sunol Tap / Rio Bravo Rocklin 100 KV
Contingency: Texon / East Bismarck 230 KV
Loading: 129.5% (Rating: 117 MVA)
DFAX: 34.7%
Pre-project loading: 105.7%
3. Facility: Lake Jackson / Terminus 100 KV
Contingency: E Bismarck / Texon Substation 230 KV
Loading: 126.0% (Rating: 128 MVA)
DFAX: 37.8%
Pre-project loading: 102.4%
4. Facility: N Charlotte / Crum Canyon 100 KV line
Contingency: Santa Teresa / Wommack 100 KV
Loading: 110.5% (Rating: 139 MVA)
DFAX: 12.9%
Pre-project loading: 103.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Terminous / Mtn Island 100 KV	Rebuild line with higher-capacity conductor	\$1,926,000	49% (\$951,000)
Sunol Tap / Rio Bravo Rocklin 100 KV	Rebuild line with higher-capacity conductor	\$2,819,000	32% (\$908,000)
Lake Jackson / Terminous 100 KV	Rebuild line with higher-capacity conductor	\$1,399,000	49% (\$691,000)
Crum Canyon Substation / North Charlotte 100 KV line	Rebuild line with higher-capacity conductor	\$13,776,000	6% (\$853,000)
	Total allocated to project		\$3,403,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.